

AAKHYA VERMA

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Analytical problem solver with experience evaluating AI systems, structuring complex data, and translating findings into actionable recommendations. Strong foundation in Python, SQL, research, and cross-functional leadership, with a growing focus on building useful technology products.

PROJECTS & EXPERIENCE

LLM Evaluation, Health & Nutrition Claims

June 2026

Python · Groq API (Llama 3.3 70B) · Prompt Testing · Structured Documentation

- Designed and executed a 33-prompt bulk evaluation of an LLM chatbot spanning contested science, high-stakes medical claims, and informal/code-mixed user input, scoring outputs against NIH, Cochrane, ACOG, and AHA sources.
- Identified inconsistent confidence calibration across similar-difficulty claims and a recurring format-compliance failure under topic complexity patterns not detectable through simple accuracy checks alone.
- Built a Python testing pipeline via API to run and log prompt batches at scale, adjusting for rate limits, to simulate a production QA/bulk-testing workflow.
- Documented findings in a structured report with failure categorization and product-facing recommendations, mirroring how AI QA findings would be handed off to an engineering team.

Dissertation Project, Bioinformatics Centre, SPPU

Aug 2025 – June 2026

GROMACS · Molecular Dynamics Simulation · Python · Structural Data Analysis

- Developed 200 ns molecular dynamics simulations of hemoglobin proteins from 4 species, generating comparative structural stability profiles to support cross-species functional interpretation.
- Identified a coupled structural transition between 120–145 ns in Glycera dibranchiata by synthesizing evidence across 5 independent structural parameters, revealing a potential oxygen dissociation event.
- Synthesized large-scale simulation outputs into organized, evidence-based conclusions and maintained reproducible documentation throughout.

Bioinformatics Intern (OJT), GeneArche, Pune

May 2025 – July 2025

Python · SQL · GWAS Catalog · dbSNP · 1000 Genomes · Data Validation

- Built a proprietary variant-phenotype database for 19 health-related traits by consolidating and cross-validating data from databases.
- Ran systematic quality-control checks to catch and resolve data inconsistencies before integration directly analogous to flagging logical gaps and inconsistencies in AI-generated outputs.
- Documented analytical workflows clearly enough to support reproducibility as new traits were added by other team members.

POSITIONS OF RESPONSIBILITY

International Academic Conference - Host & Coordinator

Feb 25–27, 2026

- Served as point of contact for speakers, delegates, and the organizing committee across a fast-moving 3-day event, resolving logistical ambiguity in real time.

Science Day Exhibition - Budget & Logistics

Feb 2025 – Feb 2026

- Owned budget and logistics for 2 consecutive years, delivering an interactive event that reached 2,000+ school students.

Human Cafe - Member to President

Sep 2021 – May 2024

- Progressed from Member to Vice President to President of this peer mental health listening club; conducted 300+ one-on-one peer support conversations and led awareness campaigns reaching the wider student body.

PR Coordinator - Enactus, Student Council & Departmental Society

2022 – 2024

- Led sponsorship outreach across 3 student organizations, approaching 100+ prospective sponsors under shifting requirements and unclear approval processes to close 30 sponsorships (~30% conversion).

EDUCATION

MSc Bioinformatics

2024 – 2026

Bioinformatics Centre, Savitribai Phule Pune University | CGPA: 8.09 — DBT Scholar (Govt. of India) | Qualified IIT JAM, TIFR GS, and GAT-B

BSc (Hons.) Biomedical Science

2021 – 2024

Shaheed Rajguru College of Applied Sciences for Women, University of Delhi | CGPA: 8.12

ADDITIONAL INFORMATION

Skills/Tools: Structured Problem-Solving · Stakeholder Engagement · Data Analysis (Python, SQL, Excel) · LLM Evaluation & Prompt Engineering · Bulk Testing & QA · Agile/Scrum Fundamentals · Research Synthesis · Data Quality & Validation · Technical Documentation · Software Testing · MS Office Suite · Canva

Extracurricular: Organized blood donation and community engagement camps (200+ participants) · Participated in debate competitions